Realizations and obstructions for the displayed patterns

Human Compression project

Abstract

We assemble the previously reviewed proofs concerning the individual patterns in the rank-three Suzuki article. Each of its three prime-case patterns has realizations with the prescribed intersections and RWPRI property. Among its ten composite-case patterns, four have impossible intersections, five have RWPRI realizations, and one has the displayed intersections but fails RWPRI. These conclusions concern the listed patterns; the source's claim that its list is exhaustive is false and is not a premise of the arguments below.

1 Coordinates and the two rigidity arguments

Throughout $q = 2^n$ with n odd. We use the classical Suzuki matrix model, its unique Bruhat coordinates, the usual subfield conjugacy classes, involution centralizers, cyclic torus centralizers and their normalizers. Prime relative degree makes an inclusion of simple Suzuki subfield groups maximal. These are the attributed structural premises of the original reviewed arguments. The source is D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63.

Write $S_d = \mathrm{Sz}(2^d)$ for the standard subgroup obtained by restricting matrix parameters to $\mathbb{F}_{2^d}$. This is initially a field-exponent index. Unique coordinates give

$$S_a \cap S_b = S_{\gcd(a,b)}.$$

In particular $S_1 = \mathrm{Sz}(2) \cong C_5 \rtimes C_4$ has order 20; it is not a simple Suzuki group. Let H_m be the unique subgroup of order m in the standard split torus, let w invert that torus, and put $D_d = H_{2^d-1} \rtimes \langle w \rangle$ for $d > 1$ dividing n. The split-normalizer uniqueness argument, also used in the accepted normalizer and subfield notes, gives

$$N_G(D_d) = D_d,$$

and exactly one standard-size subfield group of any prescribed degree containing d contains this embedded D_d.

For clarity, the argument for this uniqueness is as follows. A normalizer of D_d normalizes its characteristic odd subgroup and hence lies in the full split normalizer. A torus element h normalizes D_d exactly when $h^2 \in H_{2^d-1}$. The quotient torus has odd order, so this is equivalent to $h \in H_{2^d-1}$. Within a subfield group, every copy of D_d is conjugate to the standard one: its odd rotation group lies in a split torus, and squaring in that odd torus aligns the reflection. Pulling an inclusion back into the standard subfield, then making this internal conjugation, leaves a conjugator in $N_G(D_d)$. It therefore does not change the subfield group. This proves the asserted uniqueness.

Lemma 1. *Suppose conjugates S, T of S_a, S_b have full split dihedral subgroups lying in a common split dihedral subgroup of G, and these two dihedrals share an involution. Then S, T are simultaneously conjugate to S_a, S_b. In particular $S \cap T \cong \mathrm{Sz}(2^{\gcd(a,b)})$.*

Proof. Conjugate the common ambient split normalizer to the standard one, and then conjugate its shared reflection to w using the torus. The two rotation subgroups are the unique cyclic subgroups of their orders. The preceding uniqueness argument now forces both subfield groups to be standard. □

The involution version is equally useful. Fix $z = u(0,1)$ in the standard centre Z_q of a Sylow two-subgroup U_q. Every conjugate of S_d containing z is S_d^u for some $u \in C_G(z) = U_q$, because involutions form one class inside S_d. Its involutions commuting with z, together with the identity, are precisely Z_{2^d}: conjugation by U_q fixes the entire Z_q pointwise. Thus if conjugates of S_a, S_b share z, their common involutions commuting with z, together with the identity, form $Z_{2^{\gcd(a,b)}}$. If a, b are coprime, their intersection cannot contain a Klein four group.

2 The four impossible composite patterns

In Theorem 5.5, patterns 1, 2, 3 require distinct odd prime degrees a, b, subgroups $S \cong \mathrm{Sz}(2^a)$, $T \cong \mathrm{Sz}(2^b)$, and a split dihedral group containing their full split normalizers with a common reflection. They prescribe $S \cap T$ to have type D_{10}, C_2^2, or C_4. Lemma 1 instead forces $S \cap T = \mathrm{Sz}(2)$, of order 20. Hence all three displayed intersections are impossible. This argument does not rely on either Mersenne number being prime.

Pattern 5 requires the same coprime field degrees and an intersection $C_4 \times C_2$. That group contains a Klein four group, contradicting the preceding involution argument. This also independently excludes pattern 2. These are obstructions for every permitted relative position, not merely failures of a preferred coordinate construction.

3 The three patterns when $q - 1$ is prime

Here $q - 1$ is prime and $q \geq 8$. Put $z(x) = u(0, x)$, and $\theta = 2^{(n+1)/2}$. We use the classical facts that every nonidentity odd-order element belongs to a cyclic torus of order $q - 1, \alpha_q = q + \sqrt{2q} + 1$, or $\beta_q = q - \sqrt{2q} + 1$, with normalizer quotients of orders two, four, four. The nonidentity elements in a Sylow two-subgroup have trace zero.

Lemma 2. *The $q - 1$ matrices $wz(x)$, $x \neq 0$, represent all nonidentity odd-order conjugacy classes, one each.*

Proof. Their traces are the distinct nonzero values x^θ. An element of even order with nontrivial odd part would centralize that odd part, whose centralizer is an odd cyclic torus; this is impossible. Thus an even-order element is a two-element, and its trace is zero. Nonzero trace therefore forces odd order. The displayed matrices belong to distinct odd-order classes. Torus centralizers and normalizers give the total number of these classes as

$$\frac{q-2}{2} + \frac{\alpha_q - 1}{4} + \frac{\beta_q - 1}{4} = q - 1.$$

Indeed the normalizer quotients act freely on nonidentity torus elements, and conjugacy inside a torus is controlled by its normalizer. This proves exhaustion. In particular every prime dividing $(q-1)(q^2+1)$ occurs as $|wz(x)|$, and $(q-2)/2$ choices give order $q - 1$. □

For the first source pattern, choose involutions a, b with $|ab| = q - 1$. Their unique ovoid fixed points P, Q are distinct; otherwise they would lie in the same elementary abelian involution centre and commute. Let T be the pointwise stabilizer of P, Q and let Z_P, Z_Q be the centres at those points. Set

$$G_0 = \langle a, b \rangle, \qquad G_1 = Z_P T, \qquad G_2 = Z_Q T.$$

Then $G_1 \cap G_2 = T$. No nonidentity rotation of G_0 fixes P: its two fixed points are exchanged by a, whereas a fixes P. Two reflections fixing P would commute, impossible in an odd dihedral group. Hence $G_0 \cap G_1 = \langle a \rangle$; similarly $G_0 \cap G_2 = \langle b \rangle$, and the triple intersection is trivial.

For the second pattern, choose x with $|wz(x)| = p$, where p is the prescribed odd prime, and take

$$G_0 = H_{q-1} \rtimes \langle w \rangle, \qquad G_1 = Z_q H_{q-1}, \qquad G_2 = \langle w, z(x) \rangle.$$

The intersections are $H_{q-1}, \langle w \rangle, \langle z(x) \rangle$, and the triple is trivial. To see that the last two intersections are no larger, a subgroup of D_{2p} containing a reflection is either that subgroup of order two or the whole dihedral group. The latter would put $z(x)$ in the split normalizer or w in the point stabilizer, respectively, both impossible.

For the third pattern, the parabolics are generated by the pairs of three involutions. The two prescribed half-orders cannot both be two, by the necessary patterns proved in the preceding note. If one is two, let the other p' be odd, choose t, x with $|wz(t)| = q - 1$, $|wz(x)| = p'$ and $x \neq t$, and use $w, z(t), z(x)$. If $p' = q - 1$ there are at least three available parameters by Lemma 2. The orders are $2(q-1), 4, 2p'$. The two central involutions commute; an odd dihedral group cannot contain two distinct commuting involutions. Consequently all pair intersections are the indicated groups of order two, and the triple intersection is trivial.

For distinct odd p, p', choose x, y with $|wz(x)| = p$, $|wz(y)| = p'$. Choose λ satisfying $\lambda^{\theta+1} x^\theta = y^\theta$ and put

$$a = w, \qquad b = h(\lambda) w, \qquad c = z(x).$$

The exponent $\theta + 1$ is coprime to $q - 1$, so λ exists uniquely, and $\lambda \neq 1$. Thus $ab = h(\lambda)^{-1}$ has order $q - 1$. The traces give $|ac| = p$, $|bc| = p'$ by Lemma 2. An intersection properly containing a shared reflection would identify the corresponding prime dihedral groups, which is impossible from their different orders or would put c in the split normalizer. Thus the required intersections again hold.

For equal odd $p = p'$, choose distinct t, x with $|wz(t)| = q - 1$, $|wz(x)| = p$ and take

$$a = w, \qquad b = z(t) w z(t), \qquad c = z(x).$$

Now $ab = (wz(t))^2$ has order $q - 1$, and bc is conjugate to ac because $z(t)$ commutes with c. Also $\langle a, b \rangle = \langle w, z(t) \rangle$, while c does not belong to that dihedral group, since it is distinct from and commutes with $z(t)$. The same maximal-reflection argument gives all intersections, including $p = 5$.

In each pattern the full split normalizer is maximal in G. The reflection intersections have prime index in each odd dihedral parabolic. The torus complement is maximal in $Z_q H_{q-1}$ by its irreducible scalar action. Finally all nontrivial pair intersections over the trivial triple have prime order. These facts verify every subset in the source RWPRI axiom. We have proved existence of the displayed subgroup patterns, without inferring thinness or flag transitivity for all their possible embeddings.

4 The realizable composite patterns

For this section only, S_s denotes the standard group $\mathrm{Sz}(s)$ with a field-size index. Patterns 4 and 6 are the following one family. Let $q = s^r$, $s \geq 8$, r an odd prime, choose a prime $\ell \mid (q-1)/(s-1)$ and put $m = (s-1)\ell$. Take

$$G_0 = S_s, \qquad G_1 = H_m \rtimes \langle w \rangle, \qquad G_2 = Z_q H_m.$$

The intersections are respectively $D_{2(s-1)}, Z_s H_{s-1}, H_m$, with full intersection H_{s-1}. Unique coordinates prove these equalities. The subfield and its full split normalizer give maximal

inclusions, and the cyclic extension indices are ℓ. The complement H_m is maximal in $Z_q H_m$: its scalar field contains $\mathbb{F}_s$, cannot equal $\mathbb{F}_s$ because $m > s-1$, and has degree dividing the prime r. Thus it is $\mathbb{F}_q$, and the scalar action is irreducible. These facts verify all RWPRI subsets. The torus action on Z_q uses a power coprime to $q-1$, which does not change the generated scalar field.

For patterns 7, 8, 9 put $q = t^{ab}$, $s = t^a$, $s' = t^b$, where $t \geq 8$ and a, b are distinct odd primes. For pattern 7 choose distinct primes $p \mid (s-1)/(t-1)$, $p' \mid (s'-1)/(t-1)$ and put $m = (t-1)pp'$. Use $S_s, S_{s'}$ and $D = H_m \rtimes \langle w \rangle$. The cyclic gcd identities are

$$\gcd(m, s-1) = (t-1)p, \qquad \gcd(m, s'-1) = (t-1)p'.$$

They hold also when a selected prime divides $t-1$: its additional valuation occurs in exactly its selected relative prime degree. Equivalently, these are the two-index instances of the scalar-field arithmetic lemma in the separately reviewed split-family note. The pair intersections are $S_t, D_{2(t-1)p}, D_{2(t-1)p'}$, and the full intersection is $D_{2(t-1)}$. It is maximal in S_t, and has prime index in each larger dihedral; all subfield inclusions have prime relative degree. Hence RWPRI holds.

For pattern 8 choose a full nonsplit torus A_t of S_t, of order $m = t \pm \sqrt{2t} + 1$, and an element r of order four in its normalizer. Its ambient centralizer is one cyclic nonsplit torus A_q. Put $A_s = C_{S_s}(A_t)$ and $A_{s'} = C_{S_{s'}}(A_t)$. Their intersection is A_t, so their orders have gcd m. Choose primes p, p' with $mp \mid |A_s|$, $mp' \mid |A_{s'}|$. They are distinct, and the corresponding subgroup $C_{mpp'}$ exists in A_q. Then

$$K = \langle C_{mpp'}, r \rangle, \qquad K \cap S_s = C_{mp} \rtimes \langle r \rangle, \qquad K \cap S_{s'} = C_{mp'} \rtimes \langle r \rangle,$$

with full intersection $A_t \rtimes \langle r \rangle$. The same prime-index and maximal-normalizer checks prove RWPRI. The compatible nonsplit torus sign is determined by A_t, rather than chosen independently in the two subfields.

For pattern 10 return to exponent indices. Choose distinct prime divisors a, b, c of n, with $n/(abc) > 1$ as in the printed parameter condition. The standard subfields $S_{n/a}, S_{n/b}, S_{n/c}$ have exactly the displayed pair and triple intersections by field gcds. All one-step inclusions have prime relative degree between simple subfield groups, so every RWPRI condition holds.

5 The mixed pattern: intersections without maximality

Pattern 9 prescribes $S_s, S_{s'}$ with common S_t and a third group of type $E_q \hat{:} E_t \rtimes C_{t-1}$. Its intersections with the two simple groups have types $E_s \hat{:} E_t \rtimes C_{t-1}$ and $E_{s'} \hat{:} E_t \rtimes C_{t-1}$; the full intersection is the point stabilizer $B_t = U_t H_{t-1}$. Here the first elementary abelian factor has the literal full order q, as printed; it is not an unspecified smaller subspace.

Conjugate the common S_t to standard position. It contains its full split normalizer, so the uniqueness argument forces the two containing simple groups to be the standard $S_s, S_{s'}$. An internal conjugation in S_t puts their common Borel at B_t. The third parabolic has a normal Sylow two-subgroup P of order qt containing an elementary abelian group of order q. In an ambient Sylow two-subgroup every involution is central, so this elementary abelian group is its full centre. It consists of the identity and all involutions of P, hence is characteristic in P and normal in the third parabolic. Its unique fixed ovoid point is therefore preserved. Containment of B_t forces this point to be the standard point at infinity. Thus the normal elementary abelian group is Z_q, and containment and orders force the third parabolic and its intersections to be

$$Z_q U_t H_{t-1}, \qquad Z_s U_t H_{t-1}, \qquad Z_{s'} U_t H_{t-1}.$$

Conversely these standard groups have all the printed intersections, so the bare subgroup pattern exists.

For any $\mathbb{F}_t$-subspace V with $\mathbb{F}_s < V < \mathbb{F}_q$, the group $Z_V U_t H_{t-1}$ lies strictly between the first intersection and the third parabolic. Such a subspace exists: adjoining one vector to $\mathbb{F}_s$ gives dimension $a + 1 < ab$. The same argument gives an intermediate group above the second intersection, using $b + 1 < ab$. Hence neither remaining intersection is maximal in this parabolic. The RWPRI condition for its singleton label fails. The forced coordinates show that this is an obstruction for every realization of the printed type, not just the one displayed here.

6 What these arguments settle

The original final paragraph asks about actual existence of the displayed intersections. The retained results are

source patterns	intersections exist	RWPRI realization
Theorem 5.3, all three	yes	yes
Theorem 5.5, $1, 2, 3, 5$	no	no
Theorem 5.5, $4, 6, 7, 8, 10$	yes	yes
Theorem 5.5, 9	yes	no

These are the previously reviewed phase-five arguments, reassembled with the now accepted uniqueness and scalar-field interfaces made explicit. Their old sufficient-pattern proofs did not establish the source's completeness claim or Theorem 5.4; the present source treatment uses separate reviewed proofs and counterexamples for those assertions. No classification of all relative positions or all flag-transitive geometries is inferred from this table. The saved finite examples support the constructions but are not substituted for the uniform proofs, and their acquisitions are not repeated here.